Connecting OPNsense to an Existing Router/AP on Proxmox

How I added an **OPNsense firewall VM** in front of my homelab while keeping my home router as the WAN gateway and Wi-Fi access point.

> All IPs use placeholders like `10.0.0.x`. Substitute your own.

Goal

Insert OPNsense between the homelab and the router so:

- The **router** stays as upstream gateway and Wi-Fi AP
- **OPNsense** handles firewalling and DNS for the lab
- The **DC** keeps doing AD DS, DNS, and DHCP for domain clients

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Prereqs

- Proxmox host with **two physical NICs** (onboard + USB ethernet)
- Existing home router / AP
- OPNsense ISO (checksum-verified)
- DC VM already running

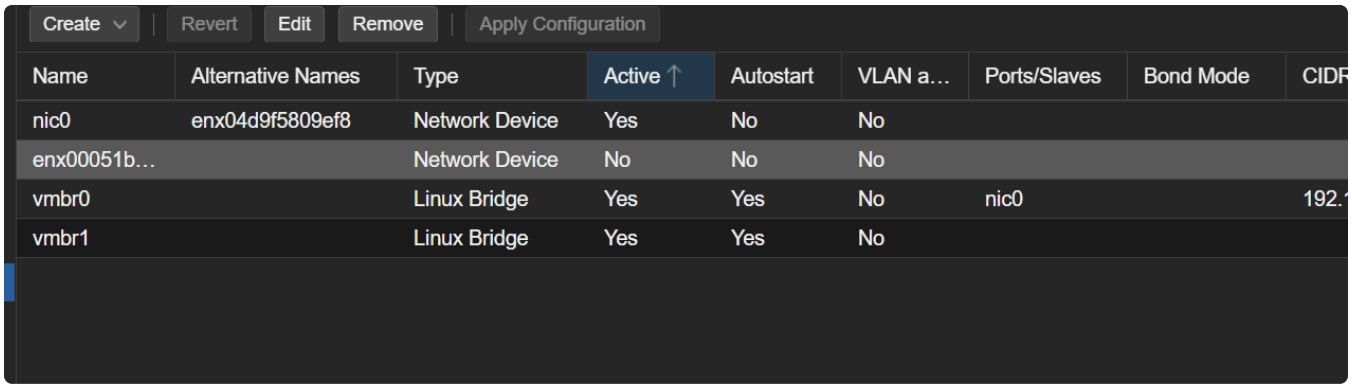
> Two physical NICs is the clean path. Single-NIC + VLANs works but needs a managed switch and is fragile.

Step 1 — Proxmox Bridges

Each OPNsense NIC needs its own bridge tied to a separate physical NIC.

BRIDGE	PHYSICAL NIC	ROLE
`vmbr0`	Onboard ethernet	LAN (homelab)
`vmbr1`	USB ethernet	WAN (to router)

If `vmbr1` doesn't exist: **Datacenter → Node → Network → Create → Linux Bridge**, attach second NIC, apply.



Create | Revert | Edit | Remove | Apply Configuration

Name	Alternative Names	Type	Active	Autostart	VLAN a…	Ports/Slaves	Bond Mode	CIDR
nic0	enx04d9f5809ef8	Network Device	Yes	No	No			
enx00051b…		Network Device	No	No	No			
vmbr0		Linux Bridge	Yes	Yes	No	nic0		192.1
vmbr1		Linux Bridge	Yes	Yes	No			

Proxmox bridges showing vmbr0 and vmbr1 attached to the two physical NICs

Step 2 — Cable It

```
[ ISP modem ] → [ Router/AP ] ── Wi-Fi to phones, laptops
                     │ LAN port
                     ▼
              USB NIC on Proxmox (OPNsense WAN, vmbr1)

              Onboard NIC on Proxmox (LAN, vmbr0)
                     │
                     ▼
              Homelab VMs (DC, Win11, etc.)
```

The USB NIC plugs into a **LAN port** on the router. That gives OPNsense's WAN a DHCP lease from upstream.

Step 3 — Create the OPNsense VM

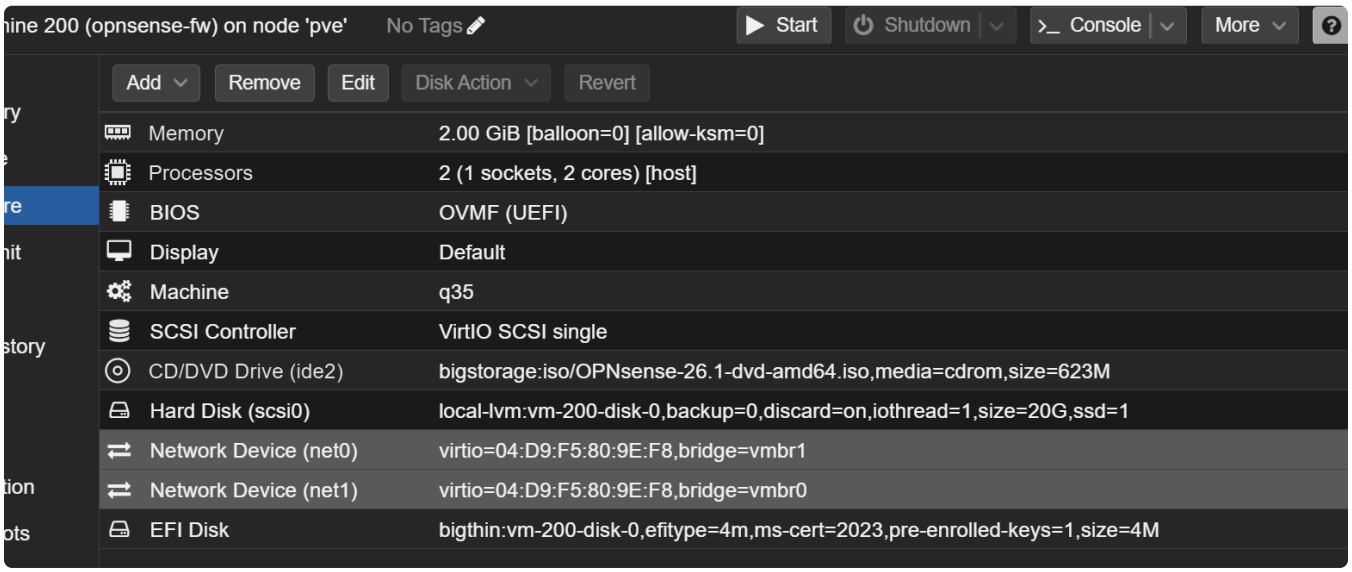
TAB	SETTING
OS	OPNsense ISO, Type: Other
Disks	32 GB, VirtIO SCSI
CPU	2 cores, type `host`
Memory	2048 MB (4096 if running Suricata)
Network	Bridge `vmbr1`, Model VirtIO, Firewall **unchecked**

Then add a second NIC (Hardware → Add → Network Device): bridge `vmbr0`, VirtIO, Firewall **unchecked**.

End state:

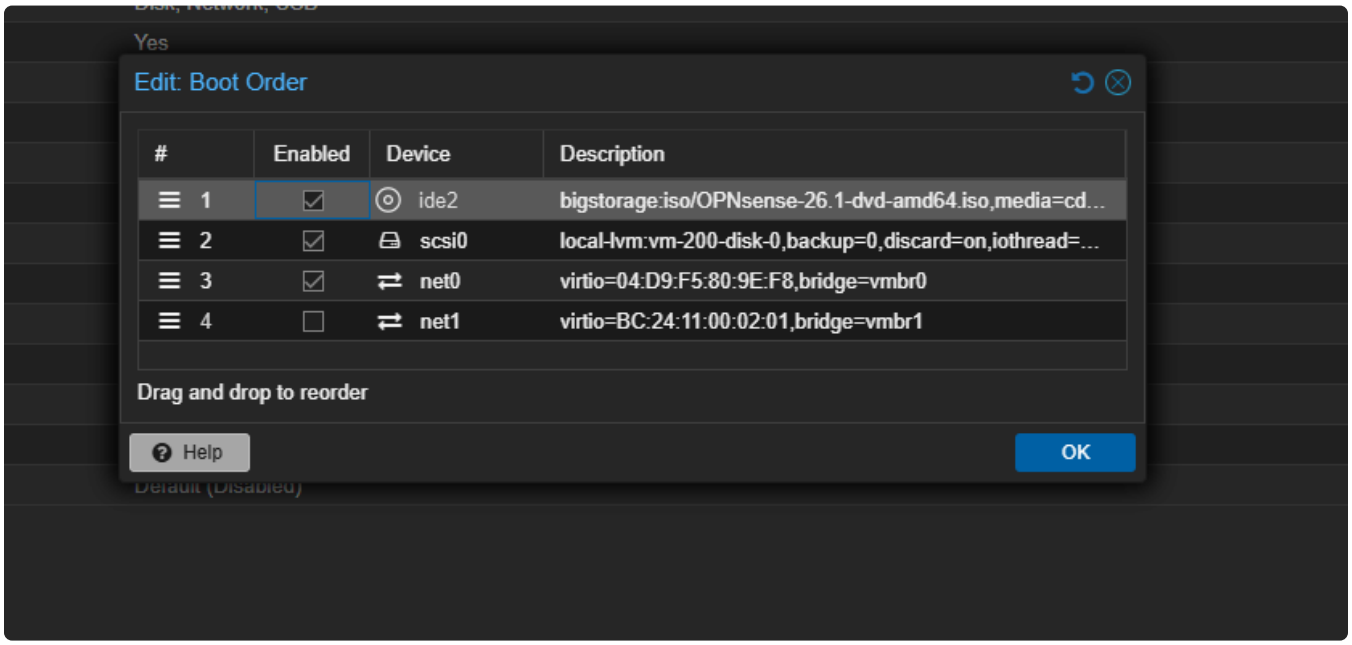
VM NIC	BRIDGE	BECOMES
`net0`	`vmbr1`	WAN (`vtnet1`)
`net1`	`vmbr0`	LAN (`vtnet0`)

> Always uncheck Proxmox's per-VM Firewall box on OPNsense NICs — it silently breaks OPNsense's own filtering.



OPNsense VM hardware configuration in Proxmox

Before first boot, make sure the ISO is set as the primary boot device:



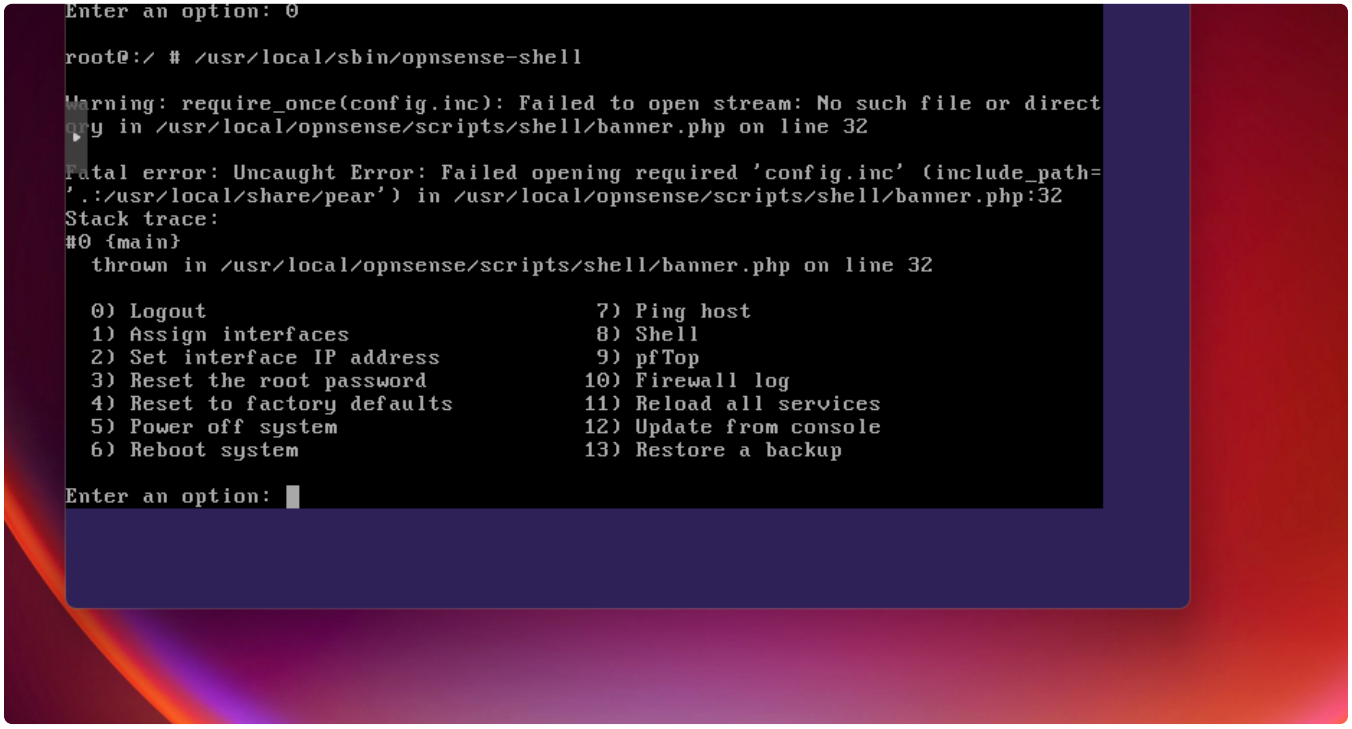
Proxmox VM boot order with OPNsense ISO first

Step 4 — Install OPNsense

Boot, install (I used ZFS), reboot, remove ISO. At the console, verify interface assignment:

- `vtnet0` → LAN (`vmbr0`)
- `vtnet1` → WAN (`vmbr1`)

If flipped, use option `1) Assign Interfaces` to swap. Getting this wrong is the #1 reason the web UI is unreachable.

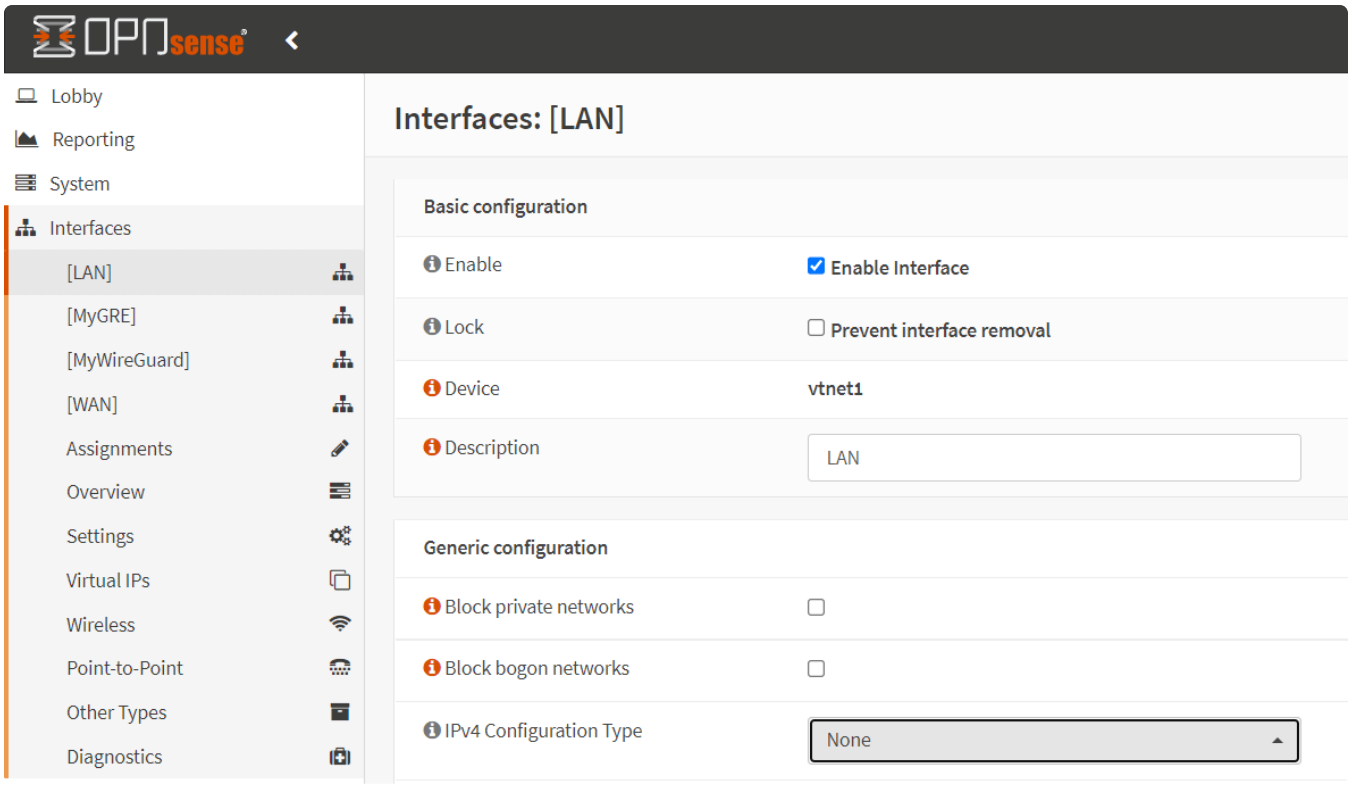


OPNsense console option menu after install

Step 5 — Configure Interfaces

LAN (`vtnet0`): static IP, e.g., `10.0.1.1/24` **WAN (`vtnet1`):** DHCP — pulls a lease from the upstream router

Browse to `https://<LAN-IP>` from a machine on `vmbr0`. Default: `root` / `opnsense`.

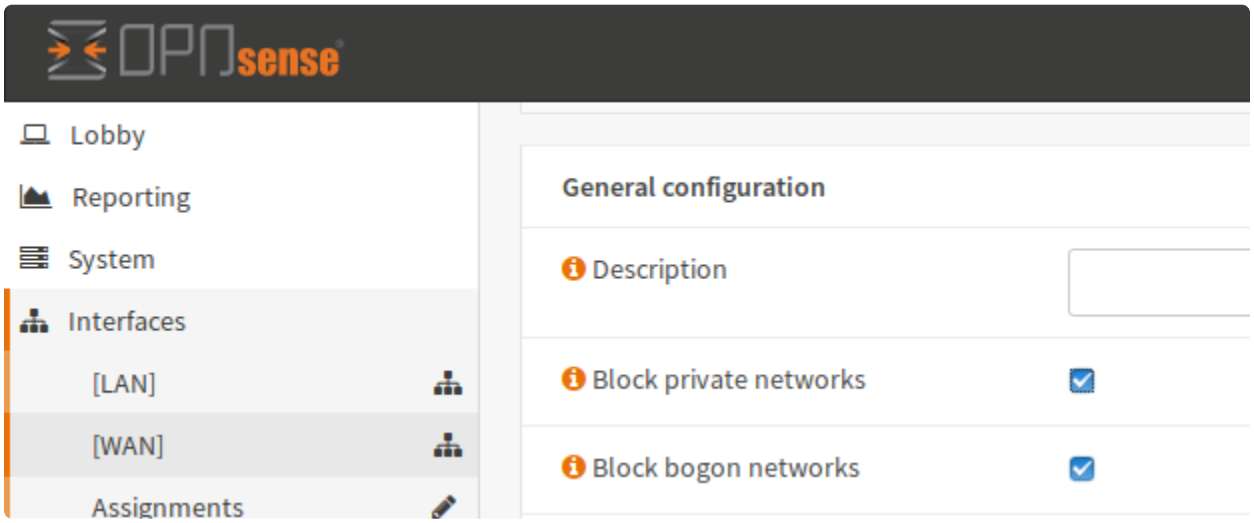


Setting the LAN interface static IP from the OPNsense console

Step 6 — Fix "Block Private Networks"

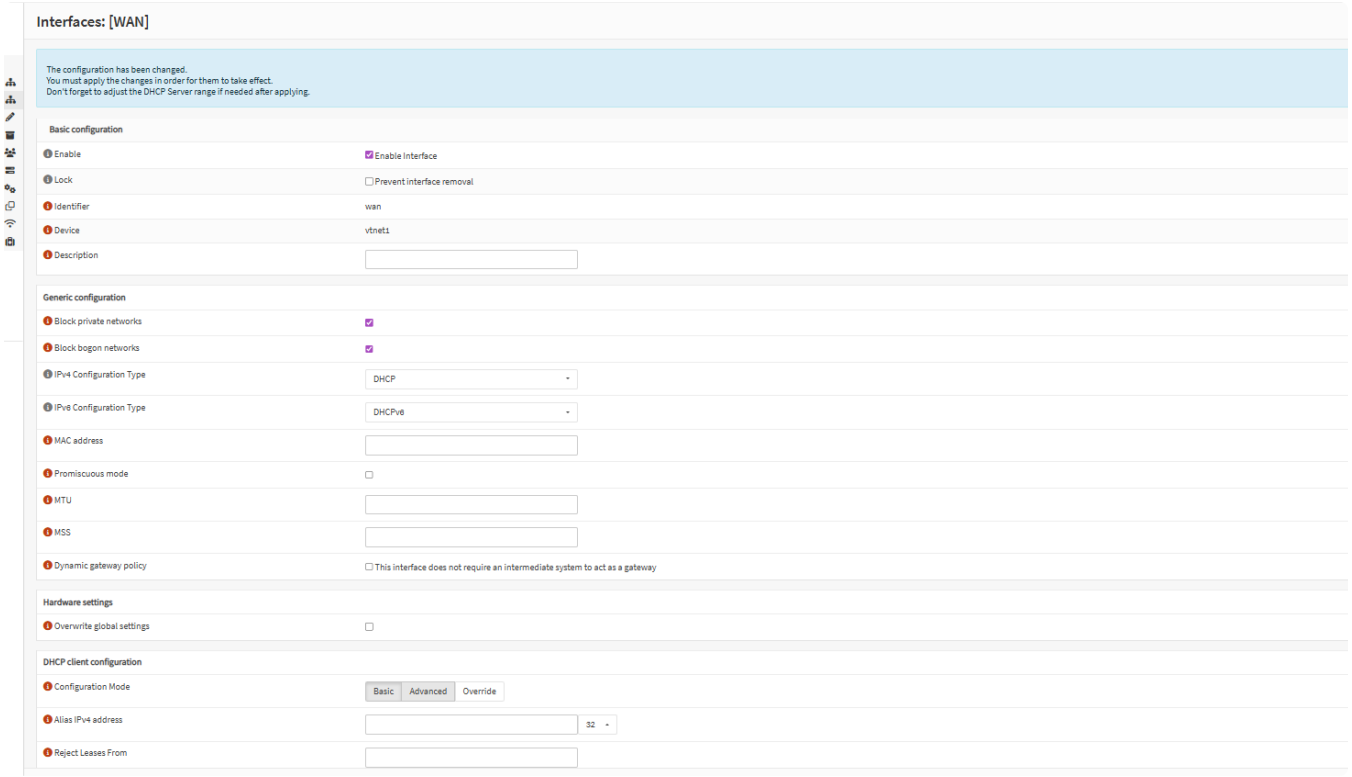
WAN defaults block private and bogon ranges. That's right when WAN faces the public internet — wrong here, since WAN faces another private network.

Interfaces → WAN, uncheck: - ☐ Block private networks - ☐ Block bogon networks



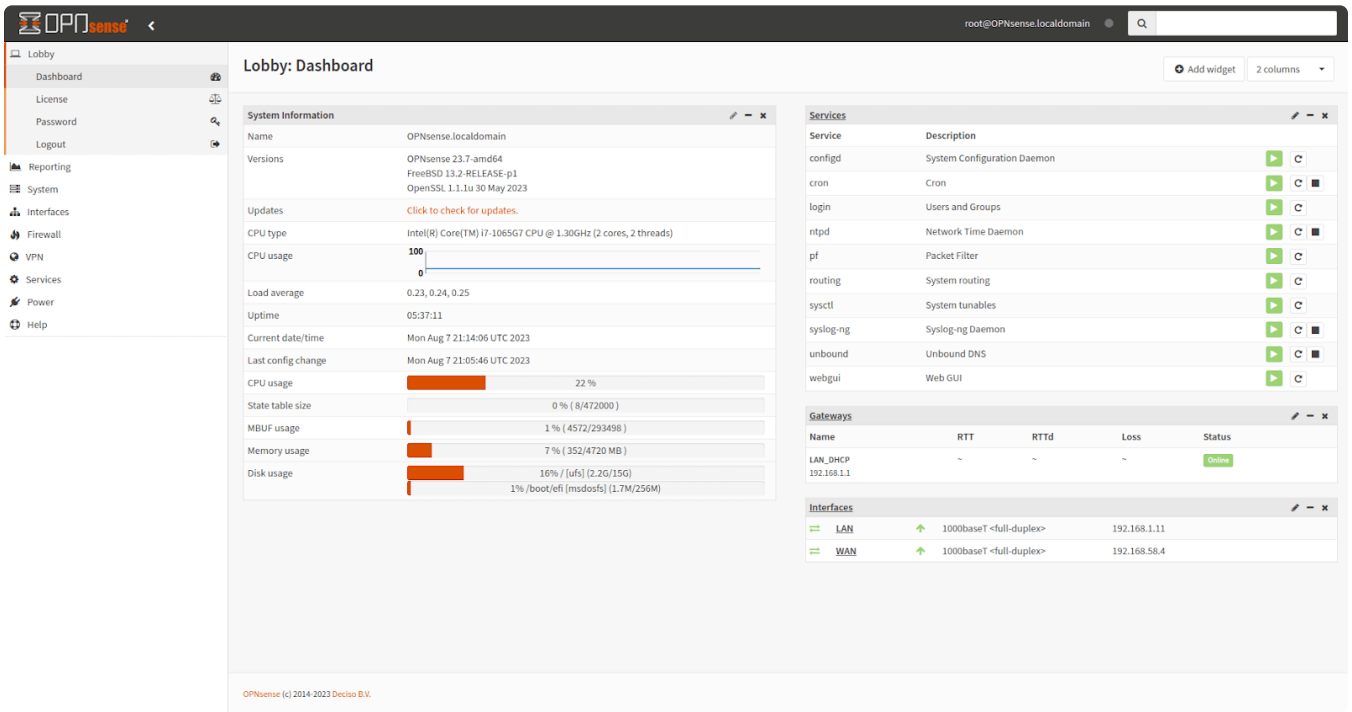
WAN interface config with the two block-private/bogon checkboxes unchecked

Save and apply. WAN should now show a DHCP lease from the upstream router:



WAN interface showing a DHCP lease from the upstream router

Dashboard view confirming both interfaces are up and routing:



OPNsense dashboard with WAN and LAN both up

Step 7 — Verify

From OPNsense console option `8) Shell`:

```
ping -c 3 <router-IP>     # should succeed
ping -c 3 8.8.8.8         # internet
ping -c 3 google.com      # DNS
```

All three pass → routing works.

```
  0) Logout                              8) Shell
  1) Assign interfaces                   9) pfTop
  2) Set interface IP address           10) Firewall log
  3) Reset the root password            11) Reload all services
  4) Reset to factory defaults          12) Update from console
  5) Power off system                   13) Restore a backup
  6) Reboot system                      14) Change keyboard layout
  7) Ping host

Enter an option:
```

Ping test reaching the upstream router

```
root@OPNsense:~ # ping -c 3 8.8.8.8
PING 8.8.8.8 (8.8.8.8): 56 data bytes
64 bytes from 8.8.8.8: icmp_seq=0 ttl=118 time=22.587 ms
64 bytes from 8.8.8.8: icmp_seq=1 ttl=118 time=22.334 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=118 time=20.747 ms

--- 8.8.8.8 ping statistics ---
3 packets transmitted, 3 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 20.747/21.889/22.587/0.814 ms
root@OPNsense:~ #
```

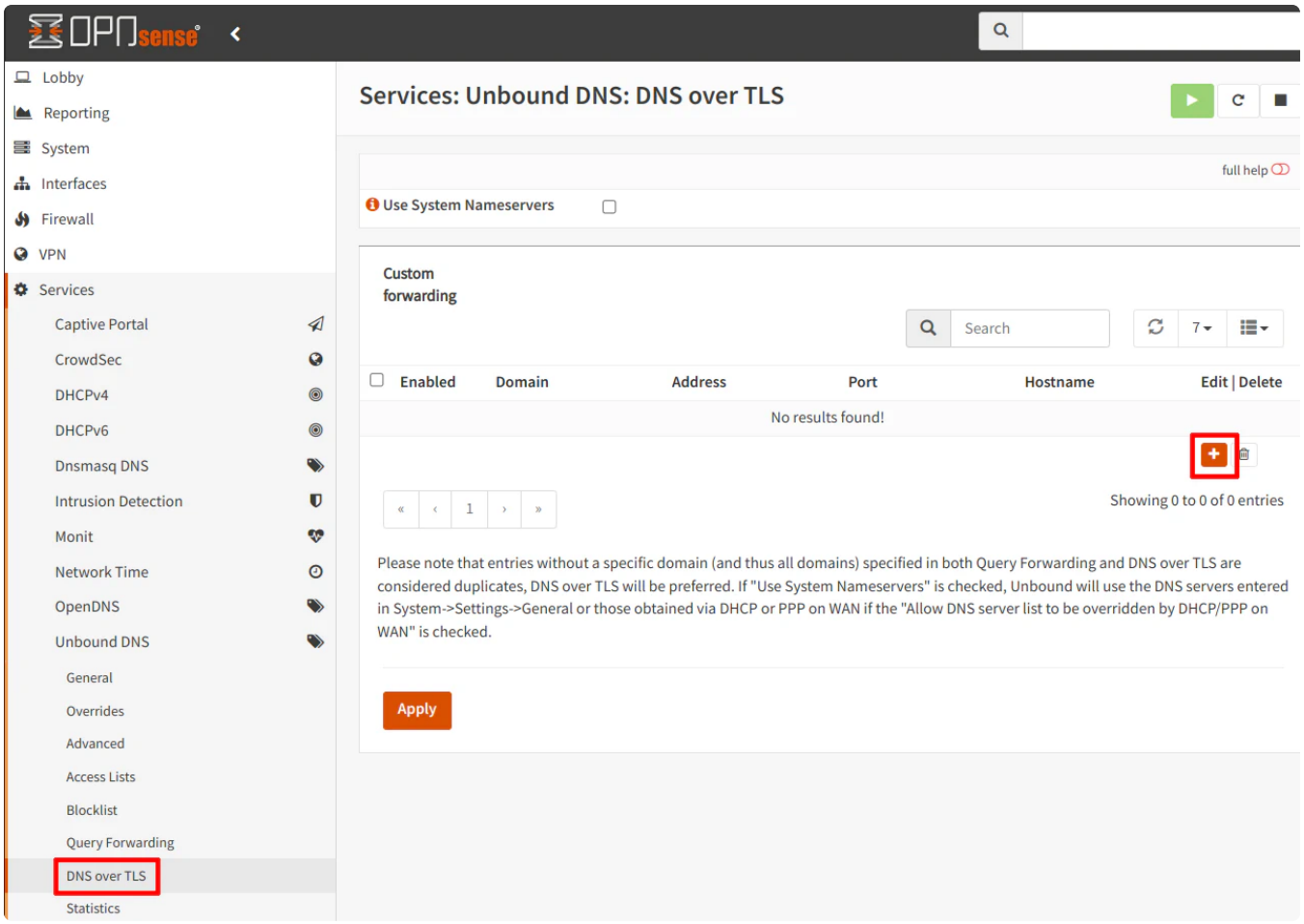
Ping test to 8.8.8.8 confirming internet reachability

Step 8 — DNS Chain

In a domain environment, the chain has to flow through the DC or AD breaks:

```
Client → DC (DNS) → OPNsense Unbound → Cloudflare 1.1.1.1 (DoT)
```

- **DC:** set DNS forwarders to OPNsense's LAN IP
- **OPNsense (Services → Unbound DNS):** enable, forward to `1.1.1.1` over DoT
- **Clients:** keep DHCP option 006 pointing at the DC



Unbound DNS configuration with DoT forwarding

Visual of the full DNS path:

Homelab DNS-over-TLS flow diagram

Homelab DNS-over-TLS flow diagram

Step 9 — DHCP

Pick one — don't run both:

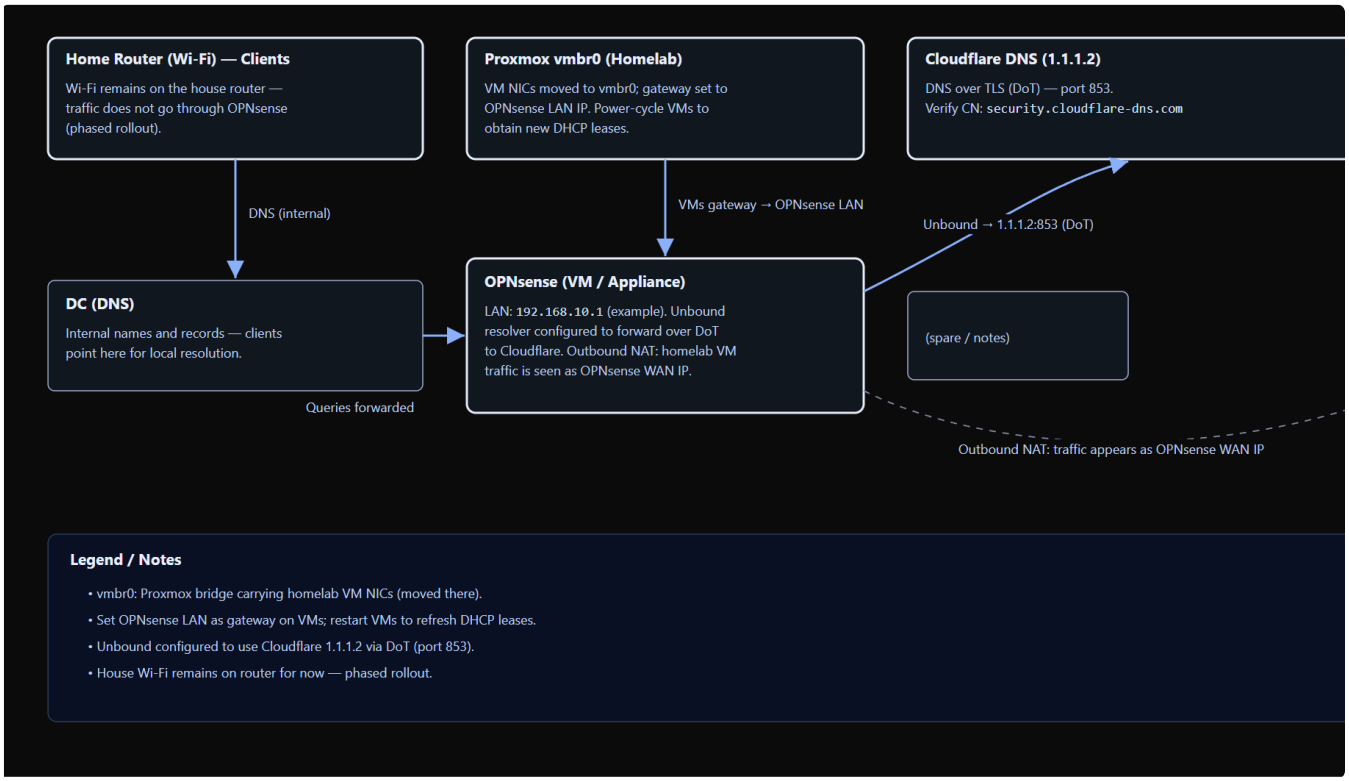
- **Keep DHCP on the DC** (what I did): disable OPNsense's LAN DHCP under **Services → ISC DHCPv4 → LAN**
- **Migrate to OPNsense:** disable the DC's DHCP role, enable OPNsense DHCP, set option 006 to the DC's IP

Step 10 — Cutover (done)

To move the homelab behind OPNsense:

1. Moved homelab VMs' NICs onto `vmbr0` only
2. Updated their gateway to OPNsense's LAN IP
3. Power-cycled clients to grab fresh leases
4. Verified outbound traffic from homelab VMs hit the router as OPNsense's WAN IP

The router still does Wi-Fi for everything else in the house — that traffic doesn't touch OPNsense, which is fine for a phased rollout.



Homelab VM successfully routing through OPNsense after cutover

Gotchas I Hit

- **Proxmox VM firewall checkbox enabled** → packets vanish. Disable on every OPNsense NIC.
- **WAN/LAN swapped at install** → can't reach the web UI. Verify with console option 1.
- **"Block private networks" left on** → WAN looks up but passes zero traffic.
- **Both NICs on the same subnet** → asymmetric routing weirdness. Long-term, put LAN on its own subnet.
- **Hyper-V default switch on a Windows VM** → hands out 172.x leases and fights your real DHCP server. Disable it on any Windows VM that doesn't need it.

Next Up

- Unbound blocklists (replaces Pi-hole / AdGuard)
- Suricata IDS in alert-only mode before flipping to IPS
- WireGuard for remote access instead of exposing RDP
- GeoIP block lists on WAN
- VLANs once a managed switch is in place